IMD3002 - 3D COMPUTER GRAPHICS

Prof. Chris Joslin

L16 – Materials: Introduction

Overview

- This lecture covers the standard reflection model, view-dependent reflection (BRDF), complex translucent layers and light scattering (subsurface scattering, SSS)

Material Properties

- Introduction
  - Under the illumination model there are three types of reflection that contribute to the final illumination of an object; these are:
    - Ambient
    - Diffuse
    - Specular
  - The final illumination of a surface or point is equal to: ambient + diffuse + specular

Material Properties

- Introduction
  - Materials reflect each component directly – we can therefore use different material coefficients
    - k_a – Ambient Reflection Coefficient
    - k_d – Diffuse Reflection Coefficient
    - k_s – Specular Reflection Coefficient

Material Properties

- Ambient Reflection
  - Ambient Light is basically background light
  - It is light that is scattered by the environment and is the Local Illumination's way of mimicking the effects of Global Illumination (i.e. it is a very simple approximation of GI)
  - Ambient light is set to be constant

Material Properties

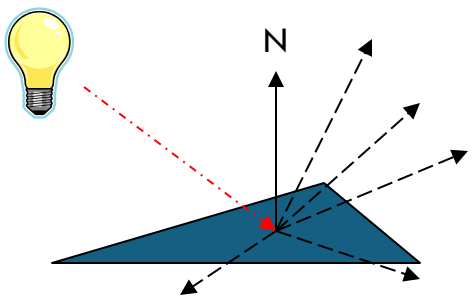
- Ambient Reflection
  - Ambient reflection (as with an ambient light) has no direction: it is independent of light position, object orientation, and the camera's position/orientation

$$A = k_a i_a$$

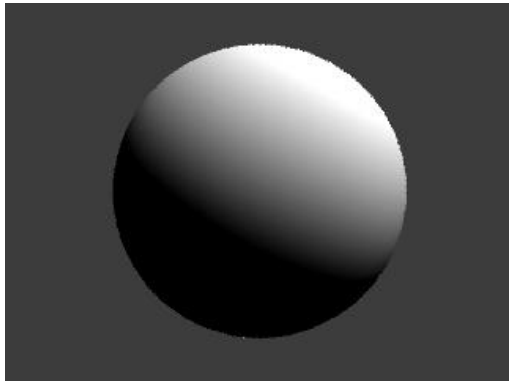
  - Where:
    - A ~ Ambient Energy
    - k_a ~ Ambient Reflection Coefficient
    - i_a ~ Ambient Light Intensity

Material Properties

- Diffuse Reflection
  - Diffuse Light is the illumination that a surface receives from a light source and reflects equally in all directions



It doesn't matter where the camera is, the same shade is produced regardless



Diffuse Reflection

Material Properties

- Diffuse Reflection
  - **Example:** objects with only diffuse light

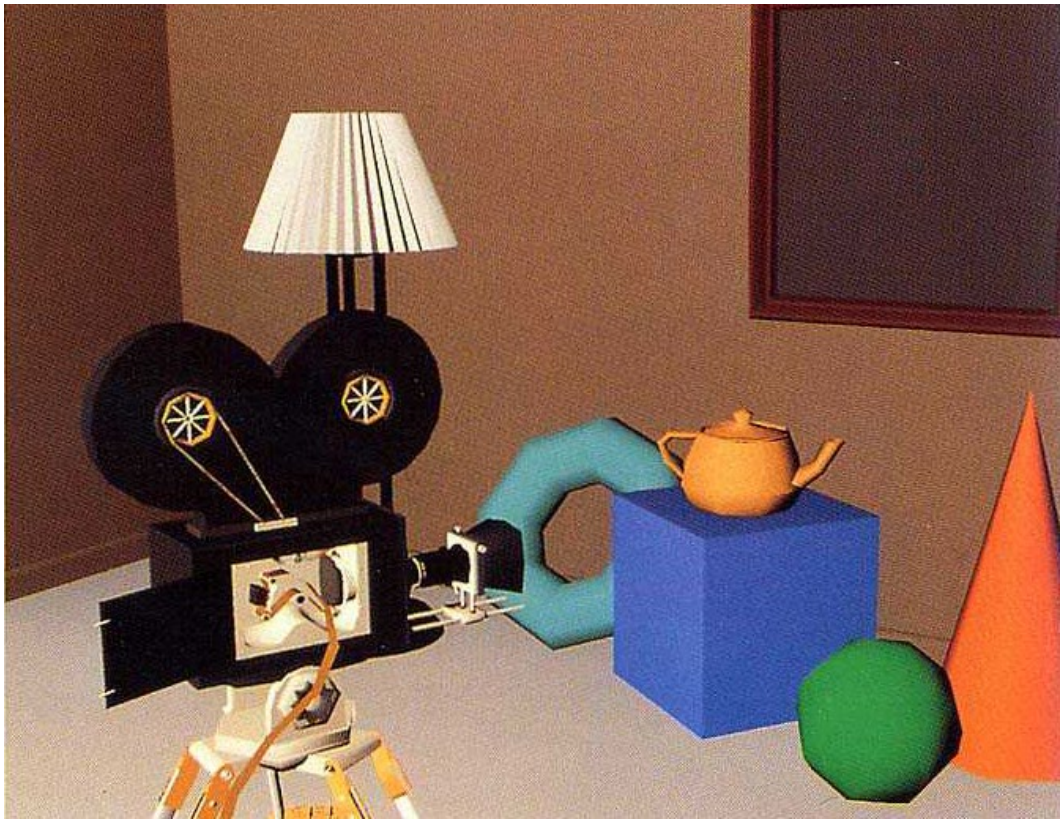
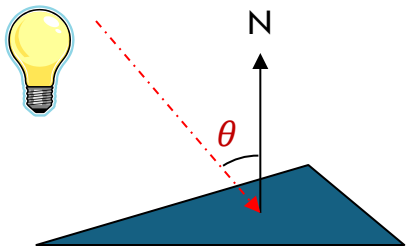


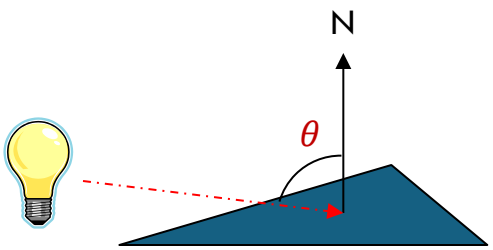
Image rendered by RenderMan (1990)

Material Properties

- Diffuse Reflection
  - The Diffuse Light calculation determines how much light the object point receives from the light source, which is based on the light source



Receives more light (θ is small)

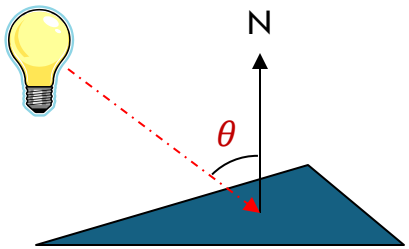


Receives less light (θ is large)

Material Properties

- Diffuse Reflection
  - Lambert's Law: the radiant energy D that the surface receives from a light source is equal to:

$$D = k_d i_d \cos(\theta)$$



  - Where:
    - D ~ Diffuse Energy (termed radiant energy)
    - i_d ~ Diffuse Light Intensity
    - k_d ~ Diffused Reflection Coefficient
    - θ ~ Angle between light vector and surface normal

Material Properties

- Diffuse Reflection
  - This equation is relatively simple, but another way to look at this is the Normal Vector (N) and the Light Vector (L)
  - The dot product between these two normalised vectors is the same as the cosine of the angle (Lecture 01); therefore the equation becomes:

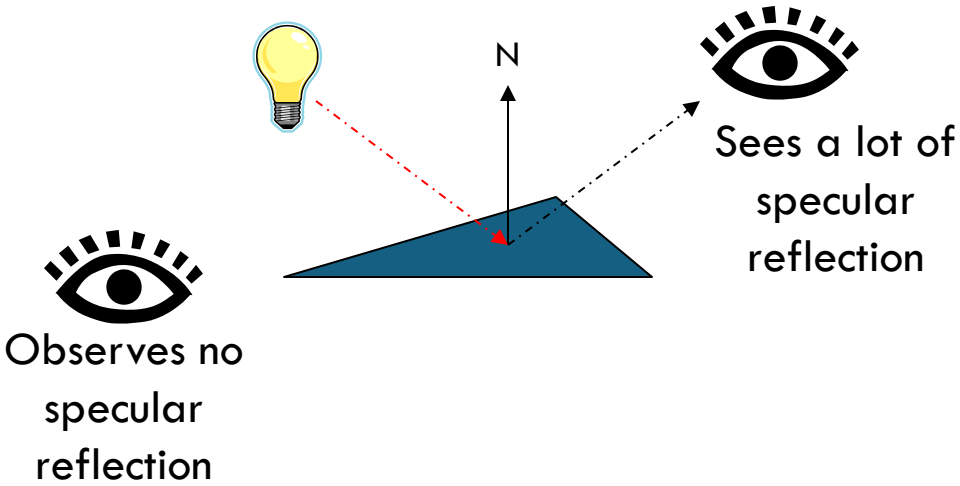
$$D = k_d i_d (N \cdot L)$$

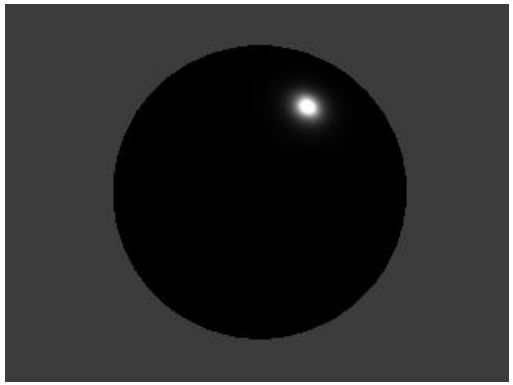
Material Properties

- Specular Reflection
  - Specular reflection is the perfect, mirror-like reflection of light from a surface, in which light from a single incoming direction is reflected into a single outgoing direction
  - Such behaviour is described by the law of reflection, which states that the direction of outgoing reflected light and the direction of incoming light make the same angle with respect to the surface normal; this is commonly stated as $\theta_i = \theta_r$

Material Properties

- Specular Reflection
  - Specular light is basically the "bright" spot on an object
  - It is the result of the total reflection of the incident light in a concentrated area





Specular Reflection

Material Properties

- Specular Reflection
  - **Example:** objects with diffuse and specular light

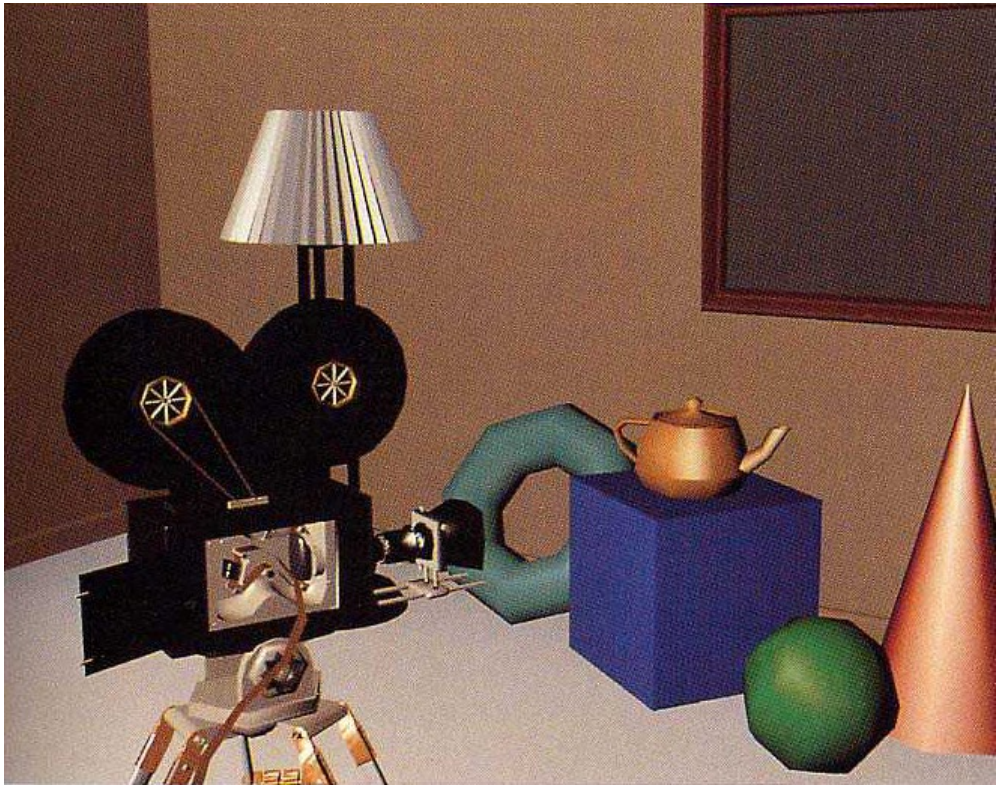
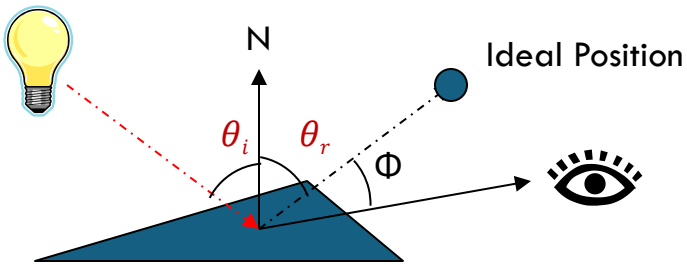


Image rendered by RenderMan (1990)

Material Properties

- Specular Reflection
  - How much reflection that can be seen depends on the position of the camera compared to the ideal position (where we obtain the mirrored reflection)



Material Properties

- Specular Reflection
  - Therefore, the specular reflection is based on the following equation:

$$S = k_s i_s cos^{\propto}(\phi)$$

  - Where:
    - S - Specular Energy
    - k_s - Specular Reflection Coefficient
    - i_s - intensity of light
    - α - Shininess Constant
    - ϕ ~ Deviation of the camera from ideal reflection angle

Material Properties

- Specular Reflection
  - The same process used for diffuse reflection (converting to vectors) can be applied to Specular
  - If the ideal reflection vector (R) is where the viewer would receive the most light, and view vector (V) is where the viewing camera actually is; and assuming both are normalised then the dot product can be applied

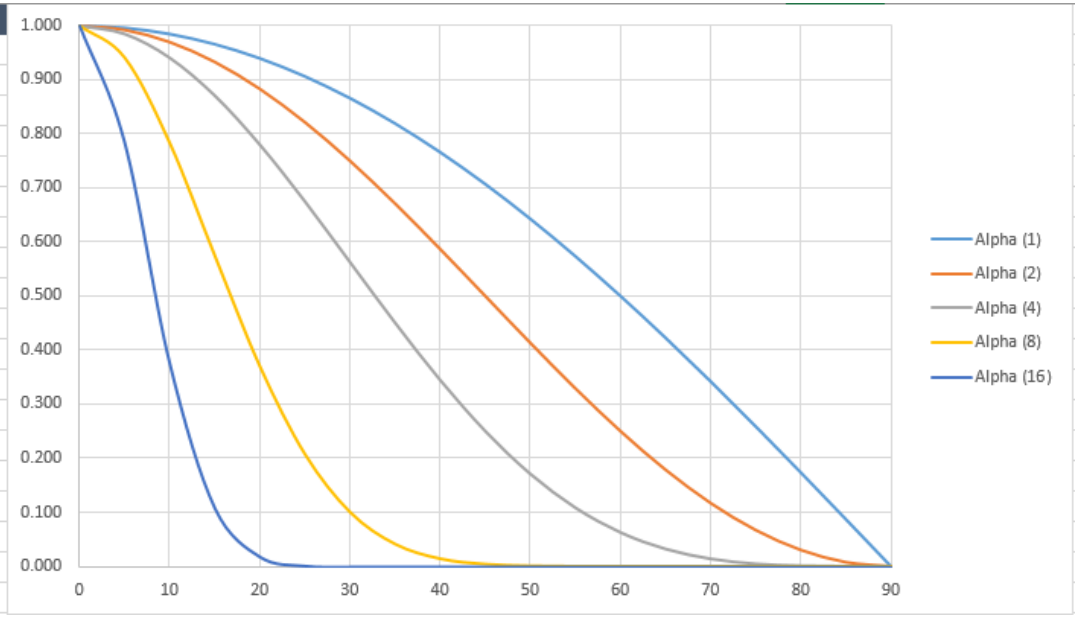
$$S = k_s i_s (V \cdot R)^{\alpha}$$

Remember: the dot product between two normalised vectors is the same as the cosine of the angle (Lecture 01)

Material Properties

- Shininess Constant
  - The shininess constant specifies how much the specular shading drops off from the ideal reflection vector
  - This can be seen from this example for different values of alpha against angles

Degrees	1	2	4	8	16 Alpha
0	1.000	1.000	1.000	1.000	1.000
5	0.996	0.992	0.985	0.941	0.783
10	0.985	0.970	0.941	0.783	0.375
15	0.966	0.933	0.871	0.574	0.109
20	0.940	0.883	0.780	0.370	0.019
25	0.906	0.821	0.675	0.207	0.002
30	0.866	0.750	0.563	0.100	0.000
35	0.819	0.671	0.450	0.041	0.000
40	0.766	0.587	0.344	0.014	0.000
45	0.707	0.500	0.250	0.004	0.000
50	0.643	0.413	0.171	0.001	0.000
55	0.574	0.329	0.108	0.000	0.000
60	0.500	0.250	0.063	0.000	0.000
65	0.423	0.179	0.032	0.000	0.000
70	0.342	0.117	0.014	0.000	0.000
75	0.259	0.067	0.004	0.000	0.000
80	0.174	0.030	0.001	0.000	0.000
85	0.087	0.008	0.000	0.000	0.000
90	0.000	0.000	0.000	0.000	0.000



Material Properties

- Reflection Models
  - Summing all these components together provides the complete (Phong) reflection illumination model

Material Properties

- Type equation here.Reflection Models
  - Here is the Phong Reflection Model equation

$$I_p = i_a k_a + \sum_{l=n}^{n} (L \cdot N) i_d k_d + (R \cdot V)^{\alpha} i_s k_s$$

  - Where:
    - I_p - Shade Value for each surface point
    - i_a, i_d, i_s - Light Intensity for Ambient, Diffuse, and Specular
    - k_a, k_d, k_s - Ambient, Diffuse, Specular Reflection Coefficients
    - α - Shininess Constant
    - N, L, V, R - Normal, Light, Viewer, Ideal Reflection Vectors
    - n, l – Number of Lights, Current Light

Material Properties

- Reflection Models
  - To add colour, we simply treat each equation with the components of Red, Green and Blue

Red

$$I_R = k_{Ra} i_{Ra} + \sum_{l=0}^{n} k_{Rd} (L \cdot N) i_{Rd} + k_{Rs} (R \cdot V)^{\alpha} i_{Rs}$$

Green

$$I_G = k_{Ga} i_{Ga} + \sum_{l=0}^{n} k_{Gd} (L \cdot N) i_{Gd} + k_{Gs} (R \cdot V)^{\alpha} i_{Gs}$$

Blue

$$I_B = k_{Ba} i_{Ba} + \sum_{l=0}^{n} k_{Bd} (L \cdot N) i_{Bd} + k_{Bs} (R \cdot V)^{\alpha} i_{Bs}$$

Material Properties

- Reflection Models
  - For common materials we actually have specific values (which actually can be determined experimentally)

Material	Ambient (Kar, Kag, Kab)	Diffuse (Kdr, Kdg, Kdb)	Specular (Ksr, Ksg, Ksb)	Shininess
Black Plastic	0.0 0.0 0.0	0.01 0.01 0.01	0.5 0.5 0.5	32.0
Brass	0.329412 0.223529 0.027451	0.780392 0.568627 0.113725	0.992157 0.941176 0.807843	27.8974
Polish Silver	0.2315 0.2315 0.2315	0.2775 0.2775 0.2775	0.773911 0.773911 0.773911	89.6

Material Properties

- Reflection Models
  - Here we have used the Phong Reflectance Model, this is a very popular model as it is fast and takes care of ambient, diffuse, and specular reflections
  - However, it is not the only model and we should be aware of the others in order to make an informed choice – the principles are very similar though and the advantages are really what sets them apart
  - The most popular models are: **Lambert**, **Phong**, **Blinn**, and **Anisotropic**

Material Properties

- Reflection Models
  - **Note:** with the exception of **Anisotropic** all models are named after their inventors
  - James Blinn is actually very famous for computer graphics and works for Microsoft
    - http://research.microsoft.com/~blinn/

Material Properties

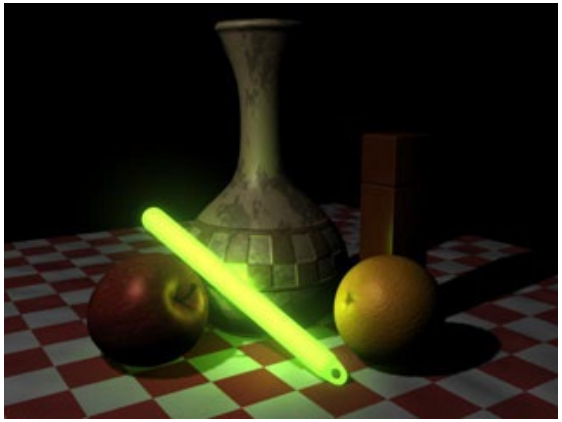
- Reflection Models
  - **Lambert Model** ~ produces a flat surface (as it lacks specular reflections and highlights) – good for modelling matte surfaces such as: chalk and matte paint
  - **Phong** ~ used to produce very shiny surfaces, good for modelling: glass and glossy plastic surfaces, or car and bathroom mouldings
  - **Phong-E** ~ similar to Phong, but provides more control over the specular highlights (additional control: roughness, highlight size, whiteness colour and specular colour)

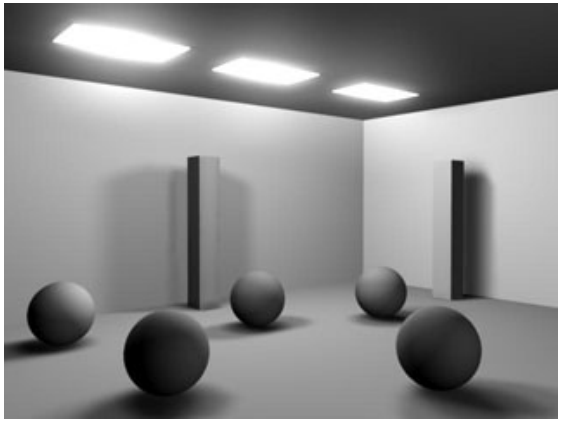
Material Properties

- Reflection Models
  - **Blinn** ~ similar to Phong, rendering takes slightly longer; good for modelling: metallic surfaces, such as brass or aluminum
  - **Anisotropic** ~ mimics the property of being directionally dependent; good for modelling: micro-grooved surfaces such as CDs, brushed metal, hair, etc

Material Properties

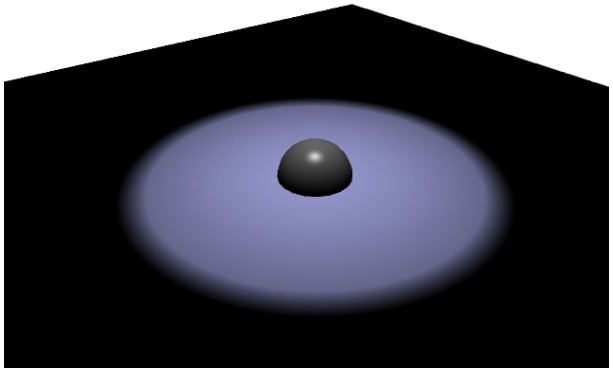
- Incandescence
  - Incandescence is the release of thermal radiation from a body due to its temperature
  - In 3D graphics this value is used to provide a light self glow and is often referred to as "**self-emission**"



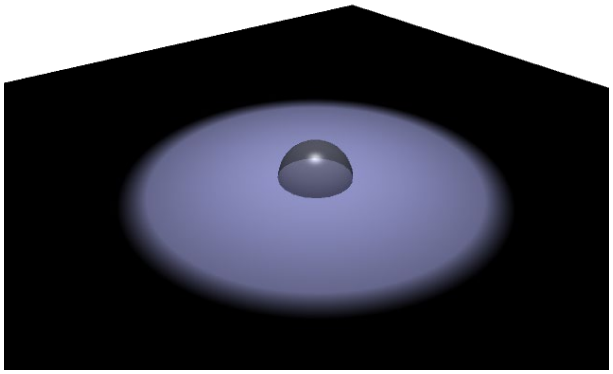


Material Properties

- Transparency
  - Transparency is factor allowing light to pass
  - Transparent materials can be seen through; that is, they allow clear images to pass
  - Transparency is not the same as “caustics”, it doesn’t refract the light at all …



0% Transparency



50% Transparency

Material Properties

- Translucence
  - Translucence simulates the way light diffusely penetrates through translucent objects
  - This means that when light shines on one side of the object, the other side is partially illuminated
  - This can be used for effects such as clouds, fur, hair, marble, jade, wax, paper, leaves, etc

Material Properties

- Translucence
  - Properties that can be set for Translucence are as follows:
    - **Translucence Ratio** ~ If set to 0.0 then no light passes through the object; if this is set to 1.0, all the light passes through
    - **Focus** ~ determines the way light scatters more in a forward direction through the object; Thin objects like a leaf, or low density objects like a cloud tend to scatter more in a forward direction. When the focus is 0.0 translucent light is scattered in all directions

Material Properties

- Translucence
  - Properties (cont'd)
    - **Depth** ~ The depth of penetration into an object at which translucence decays to zero; when the depth is set to 0.0, translucence does not decay with respect to the distance light travels through the object

BRDF

- Introduction
  - Previously we saw that the reflection properties of a material have been based on very simple lighting models (namely Phong, Blinn, etc)
  - However, despite their physical inaccuracy, they remain popular (due to good rendering times)
  - True photorealism, however, requires more sophisticated and elaborate models of surface properties

BRDF

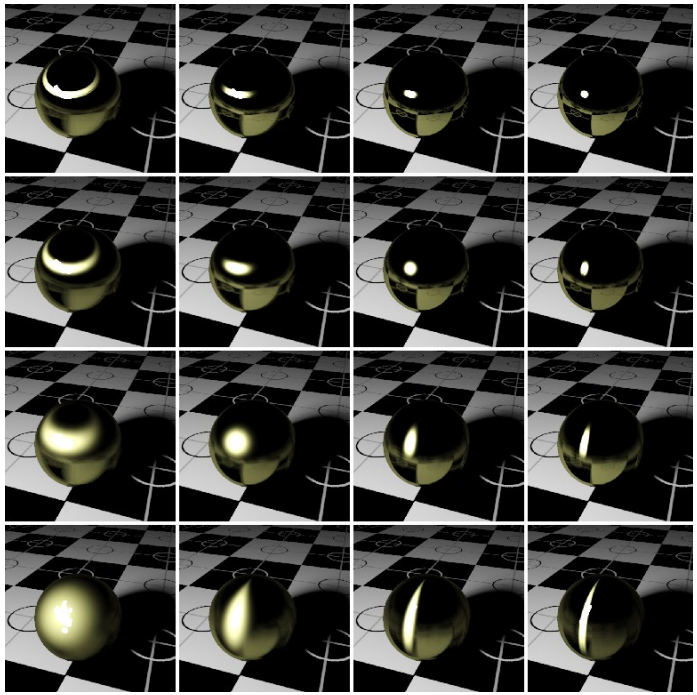
- Introduction
  - In the Phong model the term K_s is the result of assuming that the energy reflection is constant
  - Materials that have different reflection values depending on where they are viewed from change K_s into a variable
  - Therefore we end up with a different amount of energy transfer depending on where the light and the viewer are situated
    - Making K_s variable depending on direction would be visible if the camera and light were kept the same, and the object rotated (different reflections would be seen)

BRDF

- Introduction
  - When Ks is variable, then we are dealing with a Bidirectional Reflectance Distribution Function or BRDF
  - As the name implies, it is a function that describes how light is reflected from a surface
  - Therefore, it is used to describe material properties

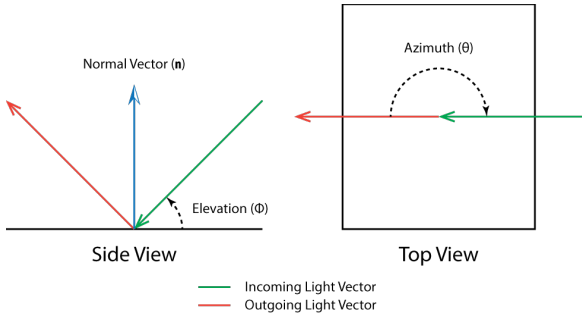
BRDF

- Introduction
  - **Example:** BRDF surfaces; we can see for each variation there is a different highlight (even though it's the same surface)



BRDF

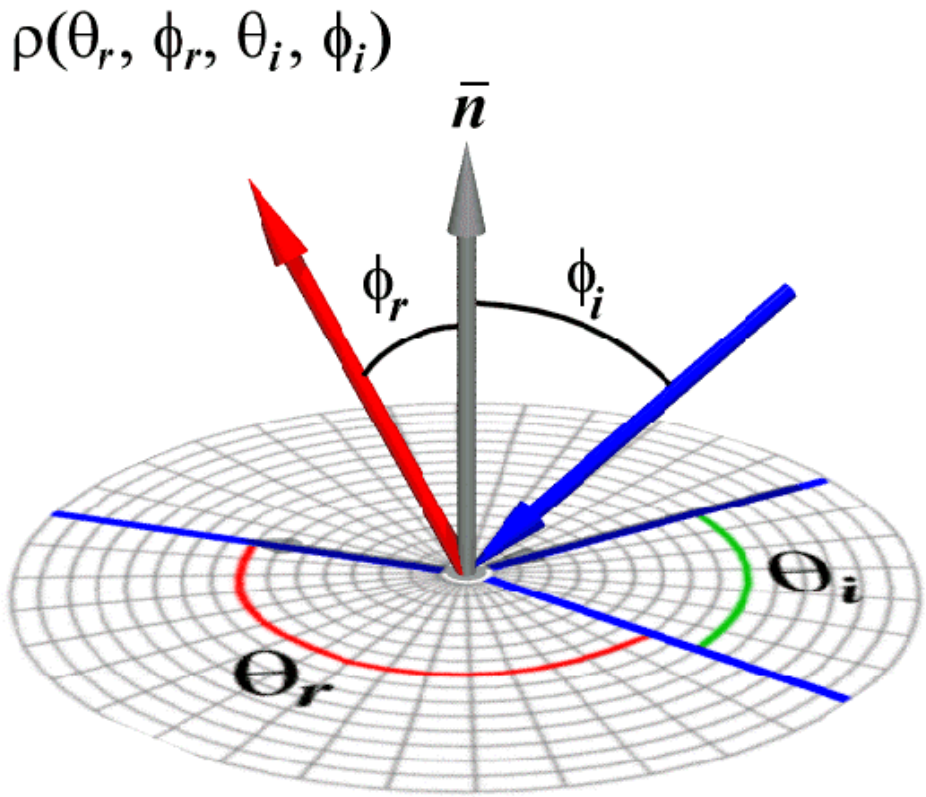
- Introduction
  - The inputs to the BRDF are the incoming and outgoing azimuth and elevation angles as measured on the surface



  - Another input is the wavelength of the light, which is represented using RGB values
    - Although there are materials that change colour depending on the incoming and outgoing directions, the general assumption is that the hue and saturation of a material remains constant

BRDF

- Introduction
  - Here is a visual representation:



BRDF

- Equation
  - The result of the BRDF is a unit-less value that is the relative amount of energy reflected in the outgoing direction, given the incoming direction

$$\boxed{L(\theta_o,\phi_o)} = \boxed{f(\theta_o,\phi_o,\theta_i,\phi_i)}\boxed{L(\theta_i,\phi_i)}\boxed{\cos(\theta_i)}$$

Output light energy for a given output angle for azimuth/elevation

BRDF Function (based on input/output angles)

Function for the light source's incoming radiance

Light scaling factor, based on incident angle

BRDF

- Equation
  - This can be simplified further by looking at the angle (azimuth/elevation) as being a vector
  - The cosine term can then be computed using the surface normal (which makes more sense)

$$L(\omega_o) = f(\omega_o, \omega_i) L(\omega_i)(n \cdot \omega_i)$$

  - Essentially what we are saying is that the light's incoming radiance $L(\omega_i)$ is modified by the BRDF function $f(\omega_o, \omega_i)$ and the angle of the surface to the light $(n \cdot \omega_i)$

BRDF

- Equation
  - Therefore, we can even look at Blinn's form of the Phong model

$$f_{phong}(\omega_o, \omega_i) = \frac{k_s (n \cdot h)^{m_{shi}}}{n \cdot \omega_i}$$

The half-angle vector (h) is from Blinn's model, and m_{shi} is the shininess factor

  - Here, we are using $(n \cdot \omega_i)$ to cancel out the one in the previous equation
  - It is a little strange that we have a term to cancel out something in the BRDF, but this occurs because it is ad-hoc and ignores the effect of the projected area

BRDF

- Properties
  - A BRDF has a few important properties:
    - "Helmholtz Reciprocity" means that the input and output angles can be switched and the function value will be the same
    - The BRDF must be normalised, which means that the total amount of outgoing energy must always be less than or equal to the incoming energy

BRDF

- The Function
  - BRDFs that are used in rendering may either be obtained from theoretical models of reflection at a surface, or can be measured directly
  - For simple shapes (spheres and cylinders) we can use just a digital camera and a stable light source
  - Adding a 3D scanner allows a wide class of curved near-convex objects to be measured
  - With measurements for a variety of materials from paints to human skin, we can achieve a high resolution and accuracy over a large range of illumination and reflection directions

BRDF

- The Function
  - We can see that the BRDF can be replaced by many different equations (or measurement sets); thus it is dependent on the material we wish to reproduce
  - The functions available are wide ranging, but they are generally complex and specific and therefore outside the scope of this course

BRDF

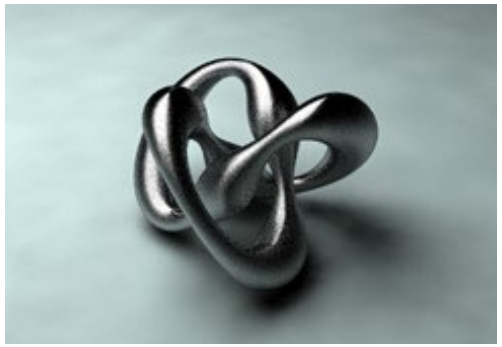
- General Function
  - It is important to note that the BRDF is a generalised function that describes how light interacts with a surface, the general equation is the Bidirectional Surface Scattering Reflectance Distribution Function
  - The BRDF describes how light comes from above a surface and leaves the surface at the same point
  - This does not include the scattering of light within the surface as seen in marble or other surfaces

BRDF

- Natural Surfaces
  - A great deal of research has been carried out for the derivation of good models for various natural surfaces
    - **Examples:** sea surfaces (clean or with various pollutions), forest canopy, various types of soil, etc., required for object classification in remote sensing
  - This has also been done for various flat but textured surfaces, for example, to simulate textures of construction materials (brick, concrete, wood, etc.)

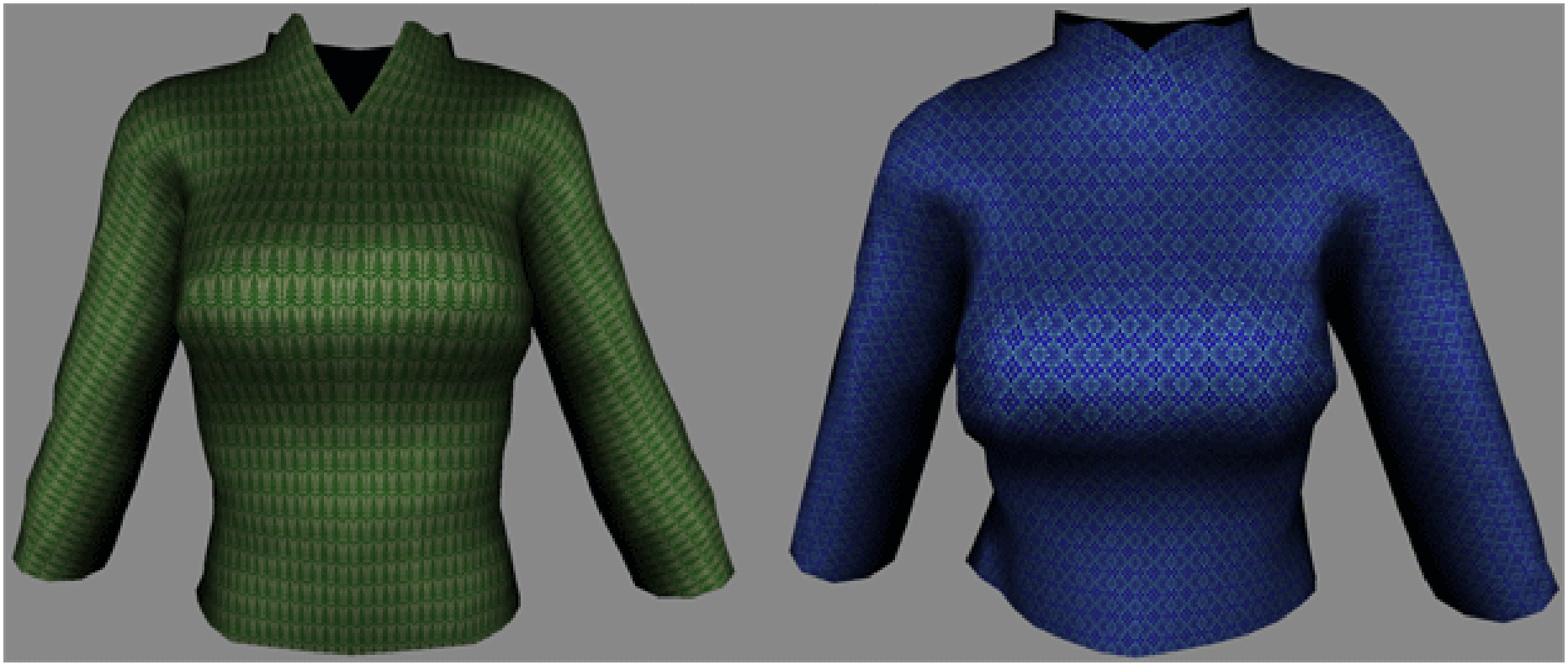
BRDF

- Natural Surfaces
  - These techniques are used for the realistic rendering in a wide range of applications, such as architectural design or video games.



BRDF

- **Example:** Cloth Weave
  - Clothing rendered using BRDFs – the weave of the cloth is used to provide a more realistic rendering



BRDF

- **Example:** Cloth Weave
  - This is the same clothing technique ….

Subsurface Scattering

- Introduction
  - Translucency is a material phenomenon where light travels penetrates an object rather than bouncing off the surface
  - Translucency is similar to transparency; however:
    - Translucency doesn't always follow Snell's Law of refraction (whereas Transparency does)
    - A translucent material scatters light inside the material before either being absorbed or exiting the material at a different location (whereas transparent materials do not scatter)
  - All non-metallic materials are translucent to some degree

Subsurface Scattering

- Lambert Reflection
  - When light hits a surface it reacts with that surface
  - Reactions include reflection, absorption, refraction, diffusion, scattering, etc
  - Dealing with a diffuse surface, especially using the ***Lambert*** model, we are just dealing with reflection
  - A light ray starts from the light source, hits a surface, a colour is determined, reflects off to the camera, and shades the appropriate pixel on the viewport

Subsurface Scattering

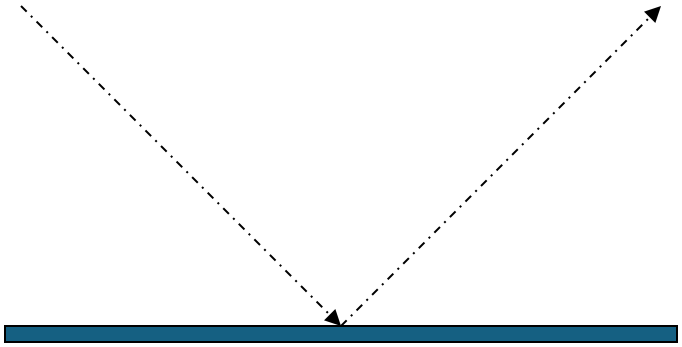
- Real Reflections
  - In the real world, this process is actually a 2-step process, light hits a surface, the surface absorbs certain wavelengths of light based on its material, and the other wavelengths are reflected back to the camera, producing the colours we see
  - The actual material weighs heavily in the equation, rough surfaces will reflect light back very differently compared to a smooth surface

Subsurface Scattering

- Real Reflections
  - The way the surface reflects light back is known as the BRDF of the surface (Bidirectional Reflectance Distribution Function)
  - BRDFs can be as basic as the Lambert Model, or as complex as the Cook-Torrance Model (which is more physically-based)

Subsurface Scattering

- Translucency
  - With a simple Lambert Model, light hits a surface and bounces, leaving the surface from the same point it strikes



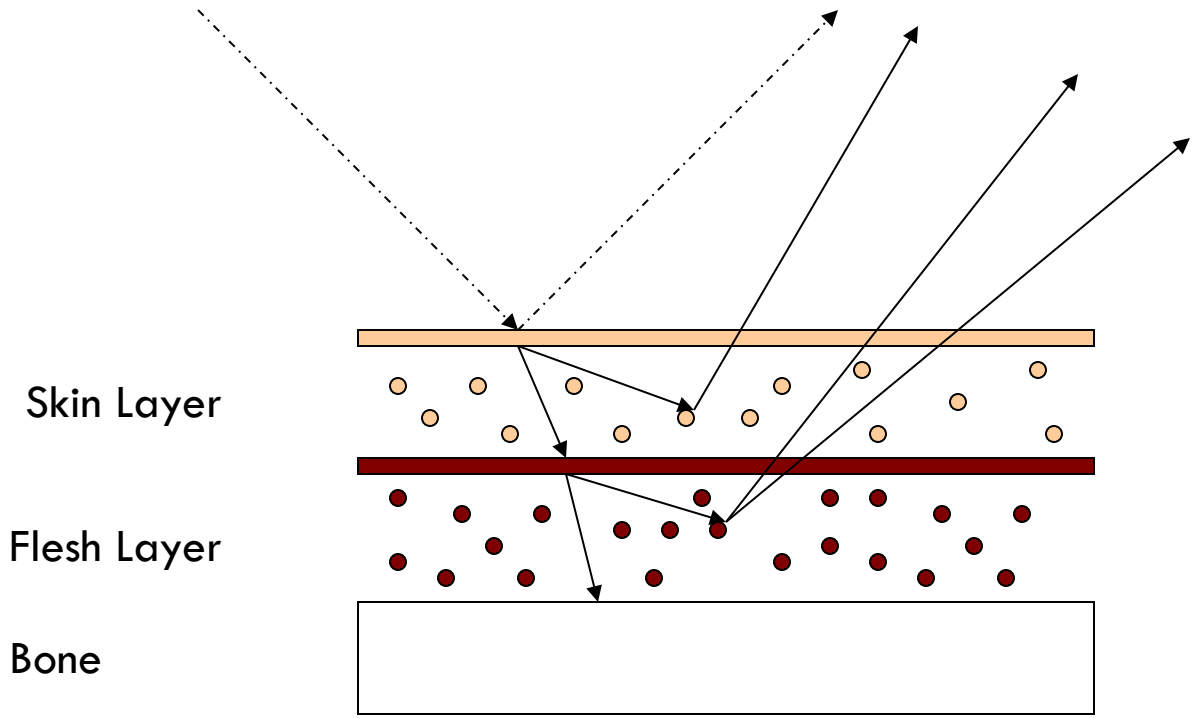
Lambert Model

Subsurface Scattering

- Translucency
  - Now let us consider a very common material: human skin
  - When light hits skin, some of the rays bounce off; some become absorbed and don't reflect; some rays enter into the skin and bounce around (and of those some become absorbed and some bounce out at a slightly different angle than they entered)
  - This is more complex than the simple parameters we have seen before as it involves different layers

Subsurface Scattering

- Translucency
  - Diagram showing light interacting with skin



Subsurface Scattering

- **Example:** subsurface scattering shader in use

Subsurface Scattering

- Producing Subsurface Scattering
  - In order to produce subsurface scattering we can use one of the following methods:
    - ***Volume Rendering*** ~ as we are not just dealing with the surface detail (such as a polygon or NURBS) model, then probably the most physically correct way of rendering SSS is to use a volume renderer which accounts not only for the surface, but also for the underlying layers

      It has the major drawback in that it is cumbersome to setup (because we need to deal with a volumetric shape), and means very heavy rendering

Subsurface Scattering

- Producing Subsurface Scattering
  - Methods (cont'd)
    - ***Kubelka-Munk Pigment Model*** ~ developed in the 1930's to describe light bouncing off wall paint; the ideal is that basically we calculate how much of which pigments we need when we paint a wall of a certain thickness in order to obtain a desired colour
      It has several drawbacks: it assumes the substance is homogeneous, and it doesn't take account the different layers or densities in a single surface

Subsurface Scattering

- Producing Subsurface Scattering
  - Methods (cont'd)
    - ***Hanrahan-Krueger Multiple Layer Model*** ~ this method allows us to solve a similar equation for multiple layers. Much like human skin, it does calculations as the light ray travels from one layer to another, to another etc – layers are therefore described by their refraction, absorption, scattering coefficients, the depth of the layer, and a phase function
    - **Note:** for more information on this, you can read up on Andrew Glassner's – "Principles of Digital Image Synthesis"

Subsurface Scattering

- Producing Subsurface Scattering
  - Methods (cont'd)
    - ***Bidirectional Subsurface Scattering Reflectance Distribution Function (BSSRDF)*** ~ we have already discussed the Bidirectional Reflectance Distribution Function (BRDF), this has been adapted for subsurface scattering
      - Previous single scattering methods don't work well with highly scattering materials (such as milk or skin) – these materials tend to scatter light in an isotropic fashion
      - Henrik Jensen's BSSDRF is a combination of diffusion and a single scattering term

Subsurface Scattering

- Producing Subsurface Scattering
  - **Example:** using Jensen's BSSDRF, for the SSS of a glass of milk

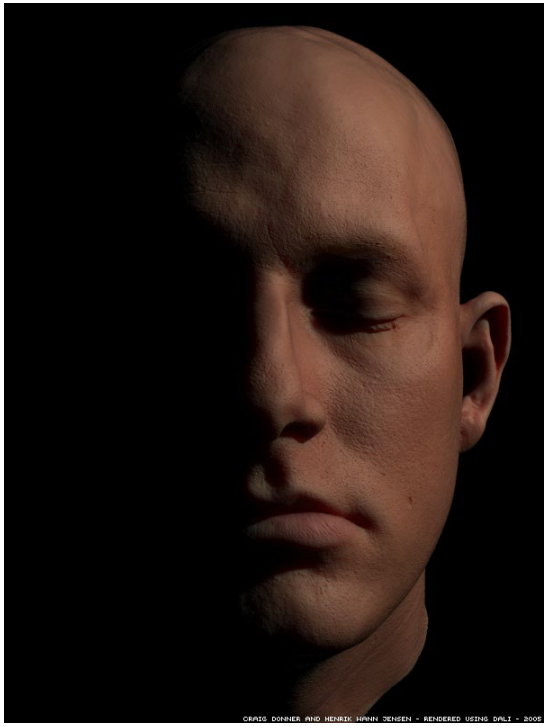
Subsurface Scattering

- Producing Subsurface Scattering
  - **Example:** marble rendered using a regular photon mapping, BRDF, and then BSSRDF (again from Jensen)

Photon Mapping	BRDF	BSSRDF

Subsurface Scattering

- Producing Subsurface Scattering
  - **Example:** a human face rendered using a multilayer model – it is very lifelike
    - This face model was rendered using a layered skin model
    - The skin is composed of three layers: epidermis, dermis, and the bloody dermis. The scattering parameters for the individual layers are from measured data available in the medical physics literature

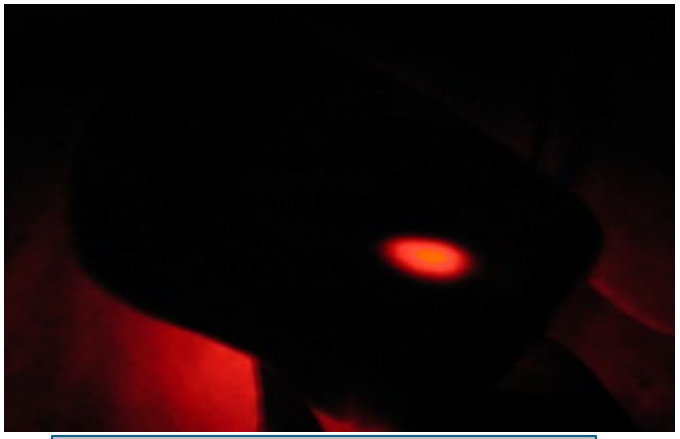


  - **Note:** the multilayered model gives the skin a less waxy look compared to the standard BSSRDF (dipole diffusion)

Subsurface Scattering

- Single Scattering vs. Multi-Scattering
  - If we have a strong focused light (such as laser) aimed at a translucent object (such as this leaf) we have a direct light – which is the single scattering, and the glow which is the multi-scattering component

Single Scattering



Multi-Scattering

Subsurface Scattering

- SSS in Maya
  - Maya supports subsurface scattering only when we used more complex renderers such as Arnold
  - Arnold generalises the subsurface method to its “standard surface” material, which allows us more control (and even to be able to turn it off)
    - There is also control in the render settings, where we can determine the sampling for SSS
    - There are AOVs (Arbitrary Output Variables) where we can output the SSS components of an image